

Poster: Blockchain-Enabled Federated Edge Learning for Big Data Quality Assessment

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Abstract—Data quality is essential to pricing big data and deciding its trading profit in digital market. Traditional machine learning-based data quality assessment methods support the valuation of data assets. Nonetheless, these methods require data to be sent over and assessed at centralized cloud, which incurs unprecedented data transmission cost and may jeopardize data privacy. To address these issues, in this poster, we propose a privacy-preserving big data quality assessment scheme (P^2QA) on the basis of blockchain and federated edge learning (FEEL). P^2QA aims to notably reduce data transmission cost, accurately measure big data quality, and effectively prevent malicious parties from violating data privacy.

Keywords—Blockchain, federated learning, data privacy, big data quality assessment, data sampling, edge server

I. INTRODUCTION

Big data has greater volume, variety, velocity, and veracity (i.e., 4Vs). It carries rich information for smart decision-making and has become a valuable commodity in digital market [2]. Businesses and organizations are keen to participate in big data trading and play different roles. They can sell their data to boost profit, broker a biparty data trading transaction, or buy data from others to conduct analytics for better strategies. With the sheer tread of big data as a core part of every business, data quality will become even more important to guarantee business success, because it measures the condition of the data in terms of completeness, accuracy, timeliness, validity, consistency, uniqueness, integrity, etc..

The big data analytics market size is estimated at 271.83 billion dollars in 2022 and is projected by Fortune Business Insights to grow up to 655.53 billion dollars in 2029. This certainly incentivizes big data trading and calls for big data quality assessment techniques to determine a legitimate price for big data. Nonetheless, big data quality assessment is a challenging issue due to the complication resulted from the 4Vs of big data. For example, data becomes outdated quickly because of high velocity, and data from social media (e.g., Facebook, Twitter, etc.) is not necessarily correct, which is one of the instances clearly defined in veracity. If big data quality gets evaluated improperly and does not reflect the condition of big data correctly, a wrong price might be advertised, and this can be a disaster for small businesses.

There has been some research on assessing the quality of big data, and traditional machine learning technique is one of the top notches [1], [3], [4]. The existing machine learning-based big data quality assessment methods require data to

be centralized on cloud for training, which causes significant data transmission overhead, let alone in the case of when persistent training is needed. In addition, sensitive users may be reluctant to participate in the training process due to data privacy concerns at the third-party cloud. Furthermore, we argue that data quality assessment should not be laboriously rely on the whole dataset when a sample of data with an acceptable level of trust can do the same. Thus, it is important to develop techniques to assess big data quality on the edge with a reasonable amount of trusted data and without violating the data privacy.

In this poster, we propose a privacy-preserving big data quality assessment scheme (P^2QA) on the basis of blockchain and federated edge learning (FEEL). The intent of P^2QA is to efficiently and effectively identify the quality of big data without violating data privacy. Specifically, data sellers are required to provide a data access interface for data broker to sample their data, the sampled data will be stored in the off-chain storage of a blockchain for latter quality assessment. The data broker informs the FEEL server to select a big data quality assessment model to be trained and send it to the edge servers in the blockchain to train the model on their local data in the off-chain storage, please refer to Fig. 1 in Section II for details. The model updates (i.e., model weights) from the training on individual edge server will be sent back to FEEL server for aggregation and a global big data quality assessment model will be generated. This global model will be saved in blockchain for trust and transparency purposes and utilized by data broker to assess big data quality and broker a transaction between data seller and buyer.

In our scheme, there is no shipment of big data between FEEL and edge servers, which significantly reduces the data transmission cost and makes big data quality assessment more efficient. we have also collaboratively polled the training results from different edge servers to assemble a more accurate model to learn the quality of big data. In addition, the data samples are derived by the data broker instead of the data seller and stored in the off-chain storage of blockchain, and we have trained the global model without shipping the sample data to FEEL server. All these guarantee the trust of the data and ease the privacy concerns of the data sellers.

The remainder of this poster is organized as follows: In Section II, we introduce the system models. In Section III, we conclude the poster and discuss future works.

II. SYSTEM MODELS

In this section, we present the models of big data trading, big data sampling, and big data quality assessment.

A. Big Data Trading Model

Big data trading typically consists of three participants (i.e., data seller, data broker, and data buyer, as shown in the top-left part of Fig. 1) and the big data transactions are completed among them [6]. In the following text, we introduce the three separate participants.

Data Seller. Data sellers (e.g., credit card and social media companies) own valuable data (e.g., full name, address of residence, email address, etc.) in their data warehouses and are willing to monetize their data assets. The data on sale is collected data from their own customers.

Data Broker. In big data trading market, different data brokers (e.g., broker for advertising and marketing and broker for financial fraud detection) manage the on-sale data from other companies, and they also collect or crawl data from public sources (e.g., government websites). The aggregated data is stored on the basis of data types (e.g., sports enthusiastic and loan applicants) to target diverse data buyers.

Data Buyer. At the end of the big data trading life cycle, data buyers (e.g., sportswear manufacturers and banks) purchase data with the help of data brokers and use it to boost, complete, or secure their businesses.

B. Big Data Sampling Model

It is important for data buyers to purchase big data from data sellers with a reasonable price, which is closely relevant to the quality of the big data. Thus, big data quality assessment by data broker is a necessity for both buyers and sellers in the first place. Nonetheless, data quality assessment on the huge amount of original dataset can be laborious and resource consuming. It would be intuitive to obtain a sample of the data and utilize the sampled data to conduct quality assessment. As shown in the bottom part of Fig. 1, in order to guarantee the trust of data, data sellers are required to provide data access interfaces for data broker to sample their datasets, and the sampled data is stored in an off-chain storage of a blockchain for latter usage. It is worth noting that data broker needs to confirm the big data sampling result with data sellers.

C. Big Data Quality Assessment Model

Data privacy has been a big concern of data sellers and they become reluctant when their data has to be processed elsewhere (e.g., centralized cloud). Data buyers, on the other hand, are worried about big data quality assessment results being altered by data brokers. Thus, we have leveraged blockchain and federated edge learning (FEEL) to tackle those issues.

Specifically, data sellers and brokers contribute part of their computing resources (e.g., an edge server) to the blockchain, as shown in the top-right part of Fig. 1. The big data quality assessment by data broker starts with the FEEL server selecting a general big data quality assessment model (e.g., a

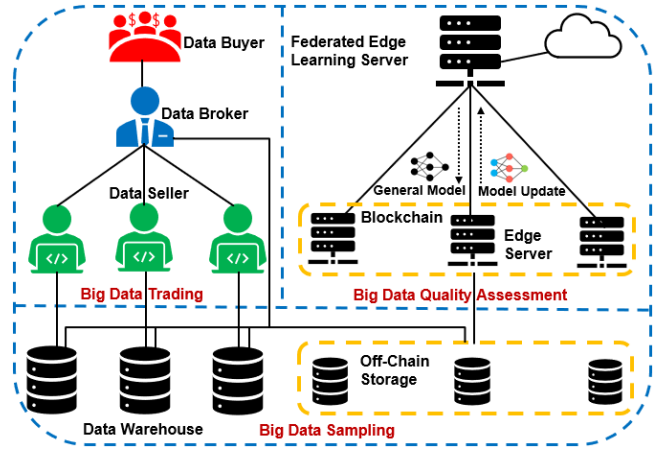


Fig. 1. Big Data Quality Assessment Model

statistical model) to be trained on the sample data in the off-chain storage. The general model is then sent to each edge server in the blockchain and is trained on the sample data locally at the edge server, and the model updates (i.e., model weights) from each edge server will be sent back to the FEEL server for collaborative aggregation. Finally, the aggregated global model will be utilized to assess big data quality. The access of the sample data and data quality assessment results can be implemented and controlled in smart contract [5] for trust and transparency among parties granted permission.

III. CONCLUSION AND FUTURE WORKS

In this poster, we proposed a privacy-preserving big data quality assessment scheme (P^2QA) to efficiently and effectively examine the quality of big data without violating its privacy. We introduced three models of big data trading, big data sampling, and big data quality assessment. In the future, we will consider the selection of the data sampling technique, the trade-off between data privacy preservation and P^2QA efficiency, and the design of an experiment to obtain evaluation results for P^2QA . This work is still at the initial stage and requires continuous efforts.

REFERENCES

- [1] D. Ardagna, C. Capiello, W. Samá, and M. Vitali. Context-aware data quality assessment for big data. *Future Generation Computer Systems*, 89:548–562, 2018.
- [2] M. Chen, S. Mao, and Y. Liu. Big data: A survey. *Mobile networks and applications*, 19(2):171–209, 2014.
- [3] K. Gu, D. Tao, J.-F. Qiao, and W. Lin. Learning a no-reference quality assessment model of enhanced images with big data. *IEEE transactions on neural networks and learning systems*, 29(4):1301–1313, 2017.
- [4] Y. Liu, Y. Wang, K. Zhou, Y. Yang, and Y. Liu. Semantic-aware data quality assessment for image big data. *Future Generation Computer Systems*, 102:53–65, 2020.
- [5] A. Vacca, A. Di Sorbo, C. A. Visaggio, and G. Canfora. A systematic literature review of blockchain and smart contract development: Techniques, tools, and open challenges. *Journal of Systems and Software*, 174:110891, 2021.
- [6] S. Zheng, L. Pan, D. Hu, M. Li, and Y. Fan. A blockchain-based trading platform for big data. In *IEEE INFOCOM 2020-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, pages 991–996. IEEE, 2020.